

Formative Assessment 3

APM1205: Linear Models

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Exercise 1: Unbiasedness of Least-Squares Estimators (10 Points)

(a) Show that $E(B) = \beta$

We are given the least-squares slope estimator $B = \sum m_i Y_i$, where the weights are defined as:

$$m_i = \frac{x_i - \bar{x}}{\sum_{j=1}^n (x_j - \bar{x})^2}$$

Under the assumption of linearity, $E(Y_i) = \alpha + \beta x_i$. Taking the expectation of B :

$$E(B) = E\left(\sum_{i=1}^n m_i Y_i\right) = \sum_{i=1}^n m_i E(Y_i) = \sum_{i=1}^n m_i (\alpha + \beta x_i) = \alpha \sum_{i=1}^n m_i + \beta \sum_{i=1}^n m_i x_i$$

Using the standard properties of the weights m_i :

- **Sum of weights:**

$$\sum_{i=1}^n m_i = \frac{\sum (x_i - \bar{x})}{\sum (x_j - \bar{x})^2} = \frac{n\bar{x} - n\bar{x}}{\sum (x_j - \bar{x})^2} = 0$$

- **Sum of weighted predictors:**

$$\sum_{i=1}^n m_i x_i = \frac{\sum (x_i - \bar{x}) x_i}{\sum (x_j - \bar{x})^2} = \frac{\sum (x_i - \bar{x})^2 + \bar{x} \sum (x_i - \bar{x})}{\sum (x_j - \bar{x})^2} = 1$$

Substituting these values into our expectation equation:

$$E(B) = \alpha(0) + \beta(1) = \beta \quad \blacksquare$$

(b) Show that A can be written as a linear function of Y_i and $E(A) = \alpha$

The intercept estimator is given by $A = \bar{Y} - B\bar{x}$. Substituting $\bar{Y} = \frac{1}{n} \sum Y_i$ and $B = \sum m_i Y_i$:

$$A = \frac{1}{n} \sum_{i=1}^n Y_i - \bar{x} \sum_{i=1}^n m_i Y_i = \sum_{i=1}^n \left(\frac{1}{n} - \bar{x} m_i \right) Y_i = \sum_{i=1}^n a_i Y_i$$

where $a_i = \frac{1}{n} - \bar{x} m_i$. Taking the expectation:

$$E(A) = E\left(\sum_{i=1}^n a_i Y_i\right) = \sum_{i=1}^n a_i E(Y_i) = \alpha \sum_{i=1}^n a_i + \beta \sum_{i=1}^n a_i x_i$$

Using the properties of the weights a_i :

- **Sum of weights:**

$$\sum_{i=1}^n a_i = \sum_{i=1}^n \left(\frac{1}{n} - \bar{x} m_i \right) = \frac{n}{n} - \bar{x}(0) = 1$$

- **Sum of weighted predictors:**

$$\sum_{i=1}^n a_i x_i = \sum_{i=1}^n \left(\frac{1}{n} - \bar{x} m_i \right) x_i = \bar{x} - \bar{x}(1) = 0$$

Thus, we find:

$$E(A) = \alpha(1) + \beta(0) = \alpha \quad \blacksquare$$

Exercise 2: Sampling Variances of A and B (10 Points)

Assuming linearity, constant variance $\text{Var}(Y_i) = \sigma^2$, and independence across observations:

1. Variance of B :

$$\text{Var}(B) = \text{Var}\left(\sum m_i Y_i\right) = \sum m_i^2 \text{Var}(Y_i) = \sigma^2 \sum m_i^2 = \frac{\sigma^2}{\sum (x_i - \bar{x})^2} \quad \blacksquare$$

2. Variance of A :

$$\text{Var}(A) = \text{Var}\left(\sum a_i Y_i\right) = \sum a_i^2 \text{Var}(Y_i) = \sigma^2 \sum a_i^2 = \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{\sum (x_i - \bar{x})^2} \right) \quad \blacksquare$$

Exercise 3: Linear Transformation of X and Y (20 Points)

(a) Transformation $X' = 10(X - 1)$

- **New slope:** $B' = \frac{1}{10}B$ due to the squared denominator scale factor in the regression slope formula.
- **Standard Error:** $\text{SE}(B') = \frac{\text{SE}(B)}{10}$.
- **Test Statistic:**

$$t'_0 = \frac{B'}{\text{SE}(B')} = \frac{\frac{1}{10}B}{\frac{1}{10}\text{SE}(B)} = \frac{B}{\text{SE}(B)} = t_0 \quad (\text{Unchanged})$$

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(b) Transformation $Y'' = 5(Y + 2)$

- **New slope:** $B'' = 5B$.
- **Standard Error:** $\text{SE}(B'') = 5\text{SE}(B)$ because the residual standard error scales proportionally.
- **Test Statistic:**

$$t''_0 = \frac{B''}{\text{SE}(B'')} = \frac{5B}{5\text{SE}(B)} = \frac{B}{\text{SE}(B)} = t_0 \quad (\text{Unchanged})$$

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(c) Effect on Hypothesis Tests and Confidence Intervals

Linear transformations of X or Y scale both the point estimator and its standard error by the same proportional factor. Consequently, hypothesis tests (and t -statistics) are completely invariant because the scaling factors cancel out entirely in the ratio. Confidence intervals scale numerically by the transformation factor, though the underlying statistical conclusions remain identical. ■